2	**A1**	Sensible choice of equipment and procedure OK (i.e. measure count rate and p.d.; change p.d. and measure new count rate) Unworkable methods/inappropriate choice of apparatus cannot score this mark	1
	A2	Voltmeter shown in parallel with the GM tube or the supply	1
	A3	Ratemeter/scalar/datalogger connected to terminals A and B of GM tube	1
	B1	Radium or Cobalt source used	1
	B2	Reason for choice Answer must relate to half-life. This mark cannot be scored if **B1 = 0**	1
	B3	Method of removing α or β radiation (depending on source used) Appropriate absorber is expected. Accept 'aluminium' or <u>thin</u> lead Could be shown on the diagram. Allow electric or magnetic deflection	1
	C1/2	Any two safety precautions e.g. use source handling tool store source in lead lined box when not in use do not point source at people/do not look directly at source Do not allow 'protective clothing', 'lead suits', 'lead gloves', 'goggles', etc.	2
	D1/2	Any good/further detail Examples of creditworthy points might be: Repeat readings (to allow for randomness of activity) or scalar + long time Sensible value of p.d. applied to GM tube (i.e. 50 V to 1000 V) Keep distance from source to GM tube <u>constant</u>/<u>fixed</u>/<u>same</u>, etc. <u>Subtract</u> count rate due to background radiation Aluminium sheets must be mm or cm thickness Allow other valid points. Any two, one mark each	2

10 marks in total

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